Concept Review

Passive Infrared Sensors

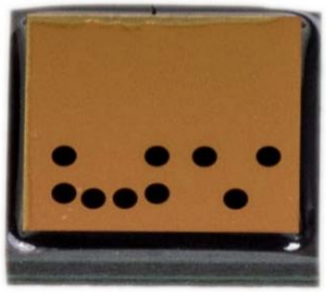
Why Do We Need Passive Infrared Sensors?

Ever entered a room and have lights turn on automatically? Or ever play a game that detects when you're stationary versus in motion? To accomplish these tasks electromechanical systems can make use of passive infrared sensors (PIR) to detect changes in infrared (heat) energy. PIR sensors fall within the family of motion sensors and can be used in a wide range of applications.

Passive Infrared Sensors

Everything radiates some amount of heat in the form of infrared light, which is invisible to human eyes but detectable through infrared sensors. Passive infrared (PIR) sensors utilize piezoelectric ceramic materials to detect changes in infrared emission.

Typically, PIR sensors combine both a sensor plat and a lens to focus the infrared emissons. Figure 1 is an example of a possible sensor and lens combination for the Kemet PL- Q873.



a. Sensor front view

b. Sensor neutral lens

Figure 1: elecromechanical elements of the Kemet PL-Q873 PIR sensor

PIR sensors are most often used to detect motion. To accomplish this, the simplest form of a PIR sensor consists of two IR measuring elements, shown in Figure 2. In an idle environment, the two elements measure a similar amount of IR. When an object moves across the detection zones, a differential signal is generated between them, producing a spike in the voltage signal.

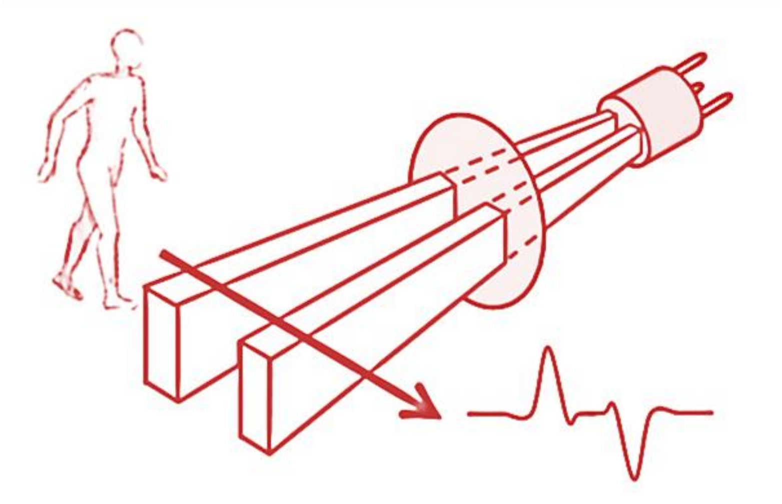


Figure 2: PIR sensors detect differential infrared levels. Figure adapted from: https://www.allaboutcircuits.com/uploads/articles/STMicro1_.jpg

Many PIR sensors are used with a lens, which can also be seen in Figure 2, to help focus incoming IR light onto the sensing elements. The lens is a critical component that determines one of the sensor's key specifications: field of view, which describes the size and shape of the detection range relative to the sensor.

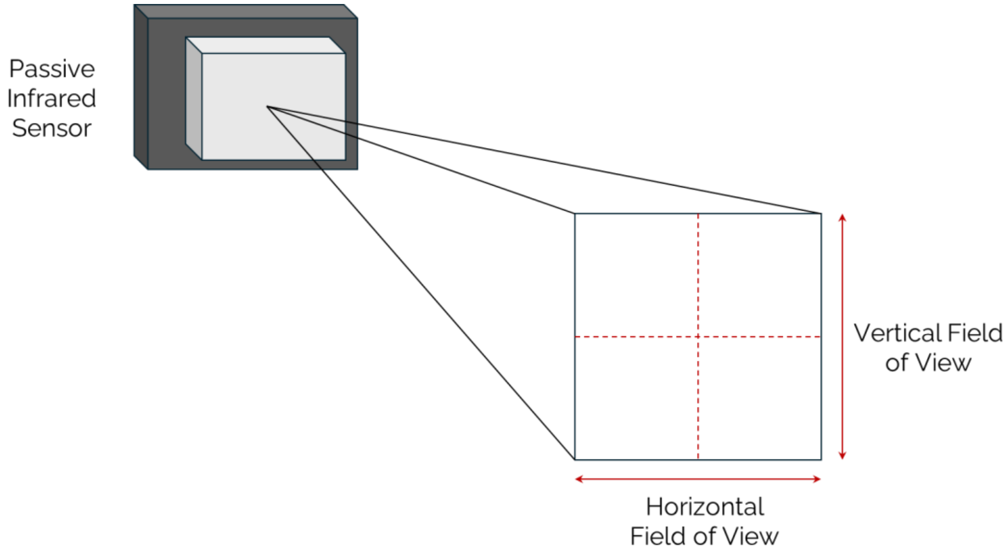


Figure 3: Representation of passive infrared sensor field of view

Sensor readings are measured in voltage and fluctuate depending if an object is moving within the PIR sensor's field of view. The figures below show sample signals measured with the PL-Q873 passive IR sensor, comparing a static environment in Figure 4 to the motion of an object crossing the field of view of the sensor in Figure 5.

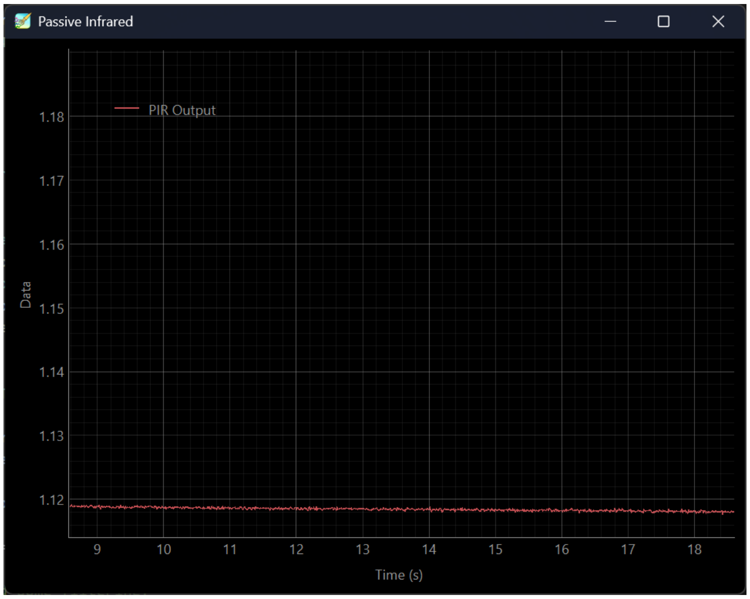


Figure 4: Sample PIR signal of static environment

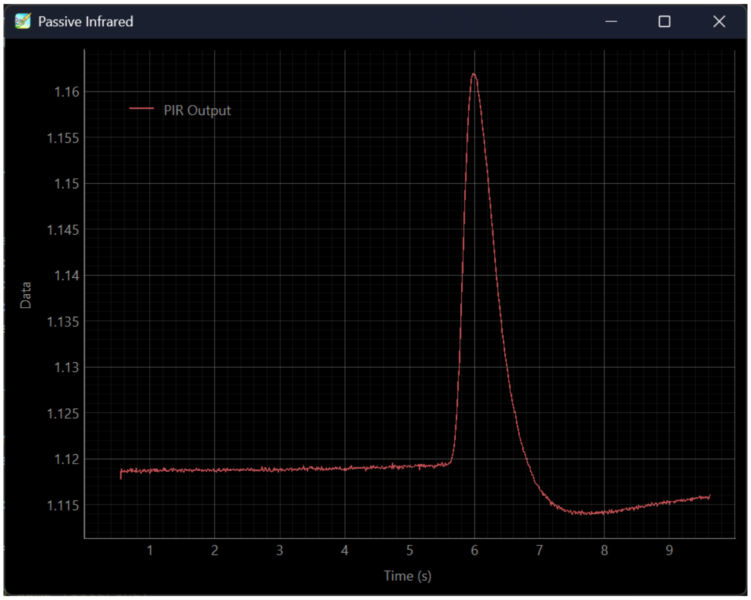


Figure 5: Sample PIR signal of motion

The rate at which the PIR sensor voltage rises and falls is related to the sensitivity of the ceramic sensor. Likewise, since PIR sensors measure infrared emission, the change in temperature between the ambient environment and the object present has an impact on how consistently a voltage spike can be detected.

Datasheet and Considerations

Selecting a passive IR sensor involves consulting the datasheet and considering a few key specifications. The requirements for sensing range and field-of-view depend largely on factors such as where the sensor will be mounted, the size of the detection target, and distance from the detection target.

Consider whether the application requires distance sensing or if simple motion detection is sufficient. Measuring the distance from the sensor with varying levels of accuracy may be necessary for performing more complex tasks such as obtacle avoidance and navigation. However, distance sensors are higher in complexity, power requirements, and cost. For applications that simply detect and react to motion, such as automated lighting, automated doors, and security systems, passive IRs are a common choice since they are inexpensive, simple, and have very low power requirements.